COOCHBEHAR GOVERNMENT ENGINEERING COLLEGE

Departmental laboratory project Of Digital communication lab(EC-592)

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Experiment -4: UNIFORM QUANTIZATION AND PCM

AIM

To implement uniform quantization and Pulse Code Modulation (PCM) encoding of a normalized sinusoidal signal, and to study the effect of different bit resolutions on quantization error and Signal-to-Quantization-Noise Ratio (SQNR).

OBJECTIVES

1. To normalize a sinusoidal signal to the amplitude range [-1, 1].
2. To implement a uniform quantizer with n quantization levels.
3. To test different bit resolutions: n = 2, 3, 4, 6 and 8 bits.
4. To generate the quantization indices for each input sample.
5. To convert the quantization indices into corresponding n-bit PCM binary codewords.
6. To reconstruct the quantized signal from the quantized levels.
7. To calculate the quantization error and Mean Squared Error (MSE).
8. To calculate and compare the experimental SQNR with the theoretical relation:
9. To observe the original and quantized waveforms and the staircase characteristic.
10. To plot the quantization-error waveform/histogram and SQNR versus number of bits.

THEORY:

1.Uniform Quantization

Quantization is the process of converting the continuous amplitude values of a sampled signal into a finite number of discrete amplitude levels.

In uniform quantization, the entire amplitude range is divided into equally spaced quantization intervals.

$$\Delta = \frac{x_{max} - x_{min}}{L}$$

For
$x_{max} = 1$
and
$x_{min} = -1$
,

$$\Delta = \frac{2}{L}$$

Therefore,

$$\Delta = \frac{2}{2^n}$$

As the number of bits increases, the number of quantization levels increases and the step size decreases.

2. Quantization Process

For every input sample
x
, the quantizer selects the nearest available quantization level.

The quantized signal can be represented as:

$$x_q = Q(x)$$

where
$Q(x)$
is the quantization operation.

The quantized waveform therefore has a **staircase-like appearance** because many input amplitudes are represented by the same discrete level.

3. Quantization Error

The difference between the original signal and its quantized version is called the **quantization error**.

$$e = x - x_q$$

For an ideal uniform quantizer, the error is approximately bounded by:

$$-\frac{\Delta}{2} \leq e \leq \frac{\Delta}{2}$$

For a sufficiently fine quantizer, the quantization error can be approximated as uniformly distributed noise.

Its mean-square value is approximately:

$$\sigma_e^2 = \frac{\Delta^2}{12}$$

Thus, increasing the number of quantization bits reduces
Δ
and consequently reduces the quantization error.

4. Mean Squared Error (MSE)

The Mean Squared Error measures the average squared differenc
between the original and quantized signals.

For
N
samples:

$$MSE = \frac{1}{N} \sum_{k=1}^{N} (x[k] - x_q[k])^2$$

A smaller MSE indicates that the quantized signal is closer to the original signal.

5. Signal-to-Quantization-Noise Ratio (SQNR)

SQNR represents the ratio between the signal power and quantization-noise power.

$$SQNR = 10\log_{10}\left(\frac{P_s}{P_e}\right)$$

For a full-scale sinusoidal signal with peak amplitude 1,

$$P_s = \frac{1}{2}$$

Using the ideal uniform quantization-noise model,

$$P_e = \frac{\Delta^2}{12}$$

Therefore, the theoretical SQNR for an ideal full-scale sinusoid is approximately:

$$SQNR \approx 6.02n + 1.76\ dB$$

This means that **each additional bit improves the theoretical SQNR by approximately 6.02 dB**.

7. Effect of Increasing the Number of Bits

Bits n	**Levels** $L = 2^n$
2	4
3	8
4	16
6	64
8	256

Expected observation

As the number of bits increases:

Number of levels ↑ → Step size ↓ → Quantization error ↓ → MSE ↓ → SQNR ↑

Therefore, an **8-bit quantizer** should produce a much more accurate representation of the original sinusoidal signal than a **2-bit quantizer**.

```
%
================================================================================
% Experiment 4 — Uniform quantization and PCM (MATLAB Implementation)
%
================================================================================
clear; clc; close all;

%% 1. Signal Generation & Parameter Setup (Task 10)
fs = 8000;          % Sampling rate (Hz)
f0 = 100;           % Signal frequency (Hz)
t = 0:1/fs:(2/f0 - 1/fs); % 2 cycles of sinusoid
x = sin(2 * pi * f0 * t); % Normalized sinusoid in [-1, 1]

bit_depths = [2, 3, 4, 6, 8]; % Task 11 bit values

% Arrays for storing evaluation metrics
mse_list = zeros(1, length(bit_depths));
sqnr_measured_list = zeros(1, length(bit_depths));
sqnr_theoretical_list = zeros(1, length(bit_depths));

fprintf('=== MANDATORY VALIDATION AND QUANTIZATION RESULTS ===\n\n');

%% 2. Execution and Validation Loop
for k = 1:length(bit_depths)
   n = bit_depths(k);

   % Perform uniform PCM quantization
```

```
    [x_q, indices, pcm_words, Delta] = uniform_pcm_quantizer(x, n, -1.0, 1.0);

    % Calculate Error Metrics
    err = x - x_q;
    mse = mean(err.^2);
    mse_list(k) = mse;

    p_signal = mean(x.^2);
    p_noise = mse;
    sqnr_measured = 10 * log10(p_signal / p_noise);
    sqnr_measured_list(k) = sqnr_measured;

    % Theoretical SQNR = 6.02*n + 1.76 dB
    sqnr_theoretical = 6.02 * n + 1.76;
    sqnr_theoretical_list(k) = sqnr_theoretical;

    % Display Results
    fprintf('Bits (n=%d, L=%d):\n', n, 2^n);
    fprintf('  Valid Range Checked : 0 <= index <= %d [PASSED]\n', 2^n - 1);
    fprintf('  Word Length Checked : %d bits [PASSED]\n', size(pcm_words, 2));
    fprintf('  MSE                 : %.6f\n', mse);
    fprintf('  Measured SQNR       : %.2f dB\n', sqnr_measured);
    fprintf('  Theoretical SQNR    : %.2f dB\n\n', sqnr_theoretical);
end

%% 3. Required Visualizations
```

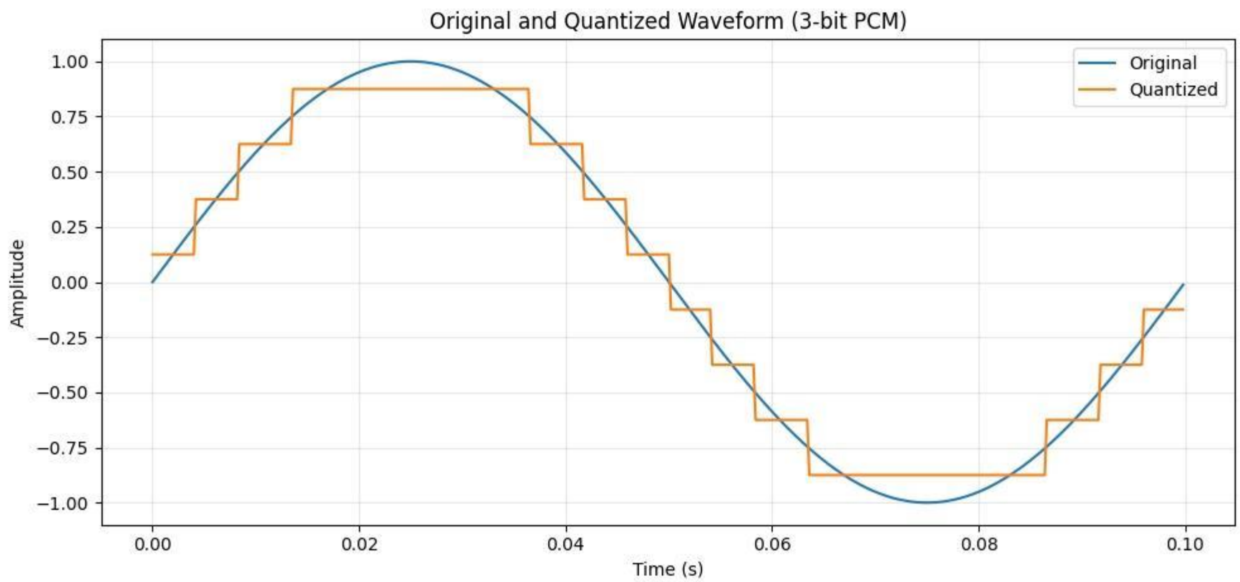
```
figure('Name', 'PCM Quantization Visualizations', 'NumberTitle', 'off');

% Viz 1: Original vs Quantized Waveform (n = 3 bits)
[x_q_3bit, ~, ~, ~] = uniform_pcm_quantizer(x, 3, -1.0, 1.0);
subplot(2, 2, 1);
plot(t * 1000, x, 'b-', 'LineWidth', 1.5); hold on;
stairs(t * 1000, x_q_3bit, 'r--', 'LineWidth', 1.2);
title('Original vs Quantized Waveform (n = 3 bits)');
xlabel('Time (ms)'); ylabel('Amplitude');
legend('Original Sinusoid', 'Quantized (n=3 bits, L=8)');
grid on; hold off;

% Viz 2: Staircase Characteristic (n = 3 bits)
x_continuous = linspace(-1.0, 1.0, 1000);
[x_q_stair, ~, ~, ~] = uniform_pcm_quantizer(x_continuous, 3, -1.0, 1.0);
subplot(2, 2, 2);
plot(x_continuous, x_q_stair, 'm-', 'LineWidth', 2);
title('Staircase Transfer Characteristic (n = 3 bits)');
xlabel('Input Amplitude x'); ylabel('Quantized Output x_q');
grid on;

% Viz 3: Error Waveform (n = 3 bits)
err_3bit = x - x_q_3bit;
subplot(2, 2, 3);
plot(t * 1000, err_3bit, 'Color', [0.9290 0.6940 0.1250], 'LineWidth', 1.2);
title('Quantization Error Waveform (n = 3 bits)');
```

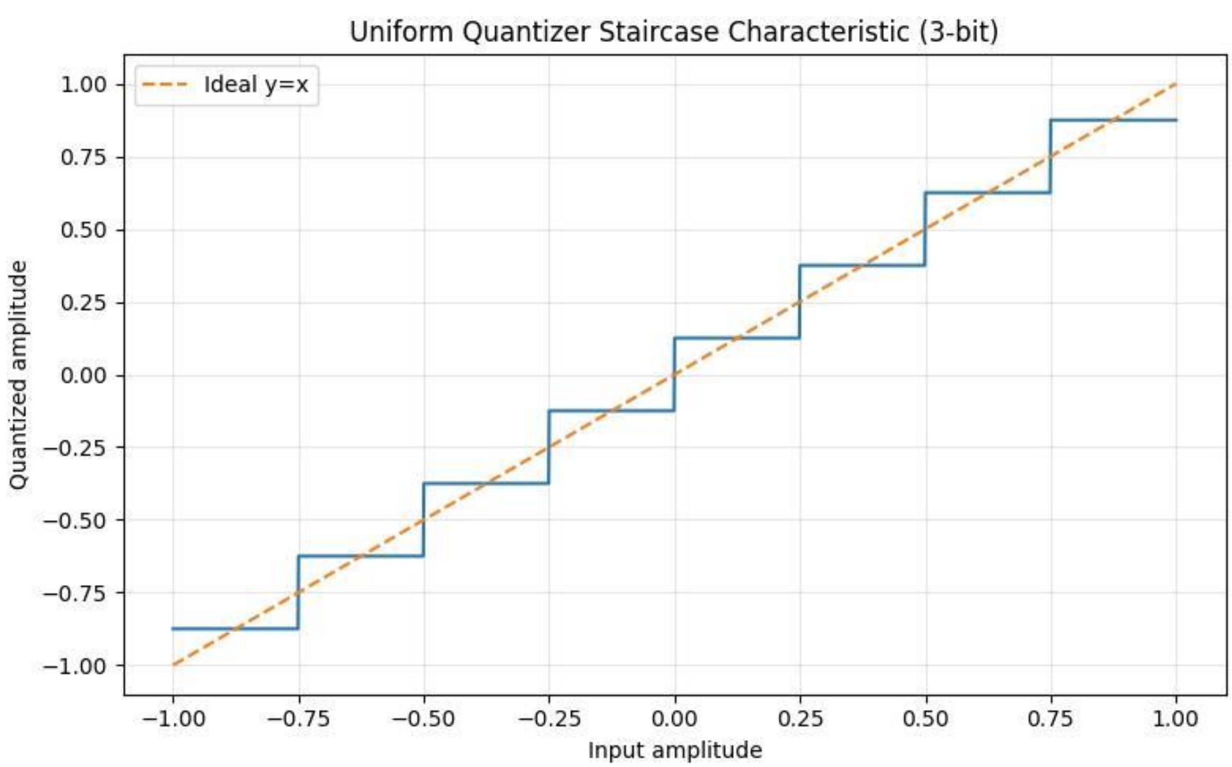
```
xlabel('Time (ms)'); ylabel('Error Amplitude');
grid on;

% Viz 4: SQNR vs Bits
all_bits = 2:8;
all_measured_sqnr = zeros(1, length(all_bits));
all_theo_sqnr = zeros(1, length(all_bits));

for i = 1:length(all_bits)
   nbits = all_bits(i);
   [xq_i, ~, ~, ~] = uniform_pcm_quantizer(x, nbits, -1.0, 1.0);
   p_sig = mean(x.^2);
   p_err = mean((x - xq_i).^2);
   all_measured_sqnr(i) = 10 * log10(p_sig / p_err);
   all_theo_sqnr(i) = 6.02 * nbits + 1.76;
end

subplot(2, 2, 4);
plot(all_bits, all_measured_sqnr, 'go-', 'LineWidth', 1.5, 'MarkerFaceColor', 'g'); hold on;
plot(all_bits, all_theo_sqnr, 'ks--', 'LineWidth', 1.5, 'MarkerFaceColor', 'k');
title('SQNR versus Bits (n)');
xlabel('Number of Bits (n)'); ylabel('SQNR (dB)');
legend('Measured SQNR', 'Theoretical SQNR (6.02n + 1.76 dB)');
grid on; hold off;
```

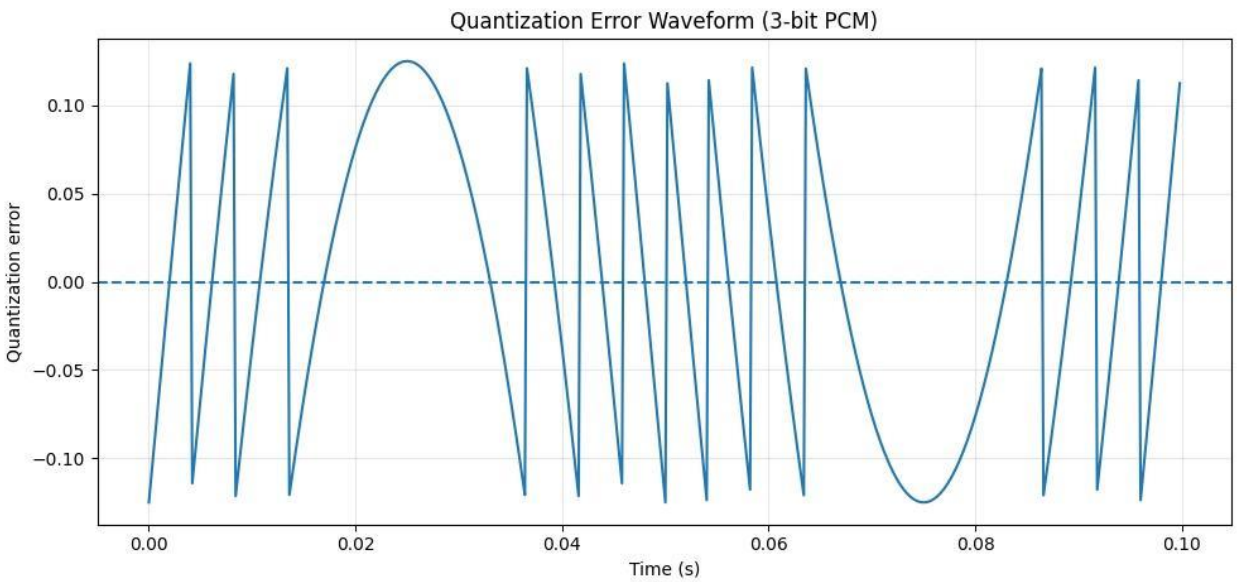
```
%%
===============================================================================
% 4. Uniform Quantizer & PCM Local Function
%
===============================================================================
function [x_q, indices, pcm_words, Delta] = uniform_pcm_quantizer(x, n, x_min, x_max)
    L = 2^n;                    % Number of levels
    Delta = (x_max - x_min) / L;  % Step size

    % Clip input to signal range bounds
    x_clipped = min(max(x, x_min), x_max);

    % Calculate bin indices [0, L-1]
    indices = floor((x_clipped - x_min) / Delta);
    indices = min(max(indices, 0), L - 1); % Ensure range boundary

    % Quantized value mapped to midpoints
    x_q = x_min + (indices + 0.5) * Delta;

    % Map indices to binary PCM word strings
    pcm_words = dec2bin(indices, n);

    % --- Mandatory Validations ---
    assert(all(indices >= 0 & indices <= L - 1), ...
        'Validation Failed: Index outside expected range [0, L-1].');
    assert(size(pcm_words, 2) == n, ...
        'Validation Failed: PCM word length does not equal n bits.');
```

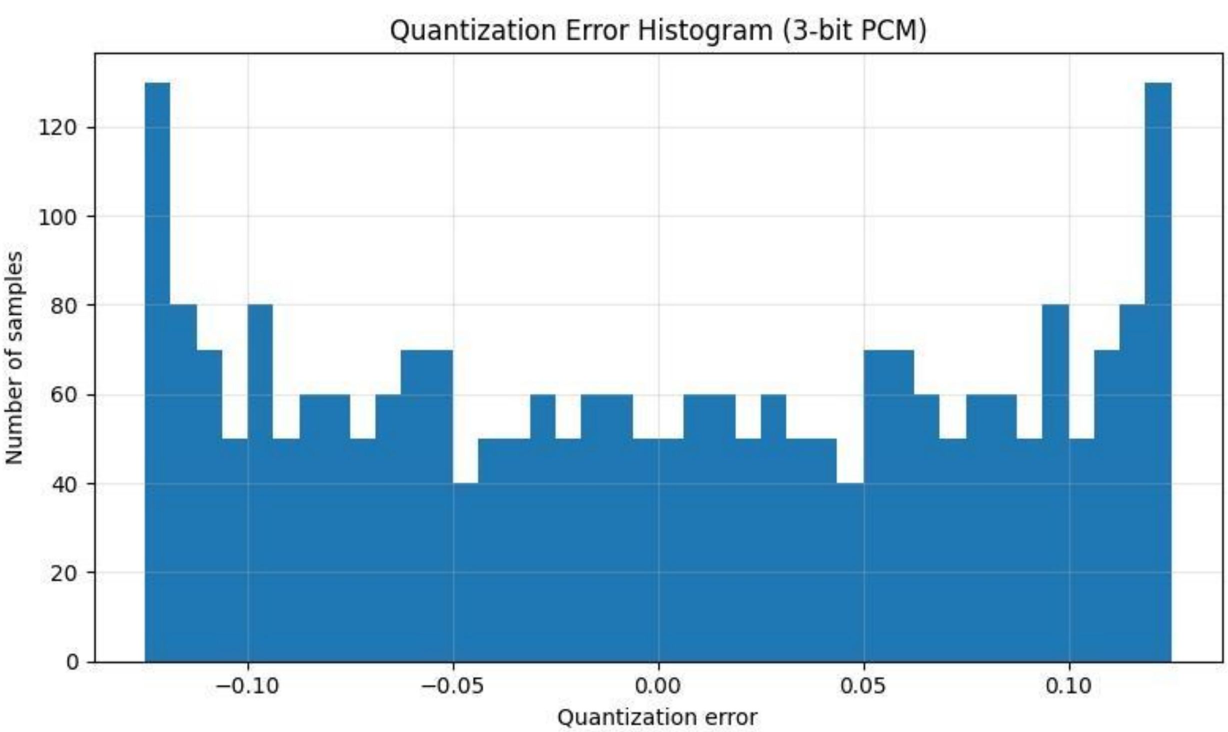
end

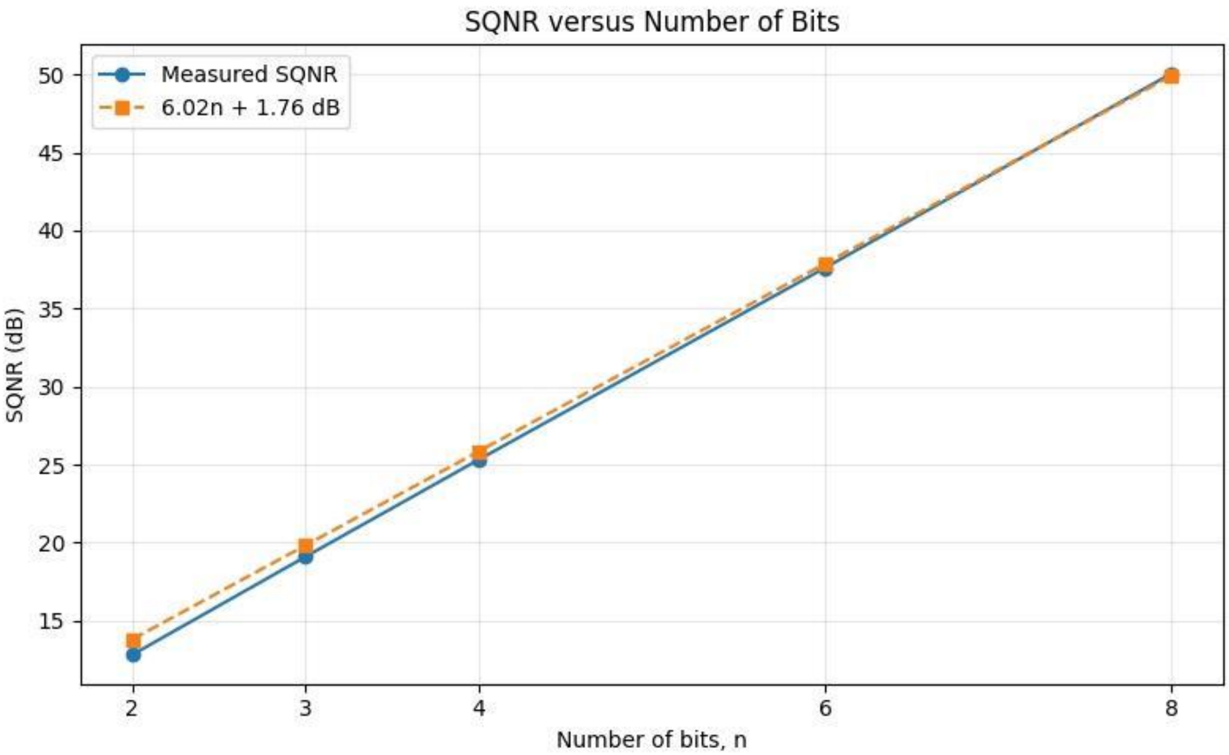
OBSERVATION AND INTERPRETATION






Quantization Error Waveform (3-bit PCM)
Quantization error
Time (s)
0.10
0.05
0.00
−0.05
−0.10
0.00
0.02
0.04
0.06
0.08
0.10


Quantization Error Histogram (3-bit PCM)
Number of samples
Quantization error
120
100
80
60
40
20
0
−0.10
−0.05
0.00
0.05
0.10



Final conclusion

Uniform PCM quantization was successfully implemented for 2, 3, 4, 6 and 8 bits. The simulation confirms that increasing the number of quantization bits increases the number of levels, reduces quantization error and MSE, and improves SQNR at approximately 6 dB per additional bit, consistent with the theoretical 6.02n+1.76 dB relationship.

Thank you..!!!